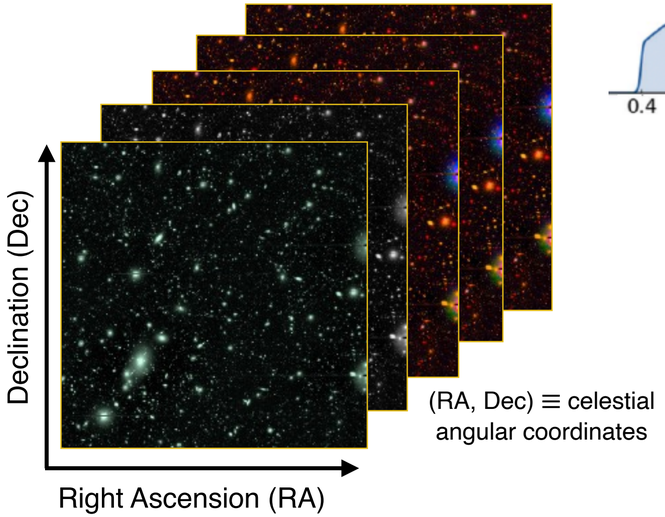
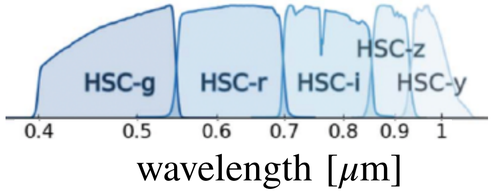
Photometric observations and galaxy models

Observations



Typical filter transmission curves

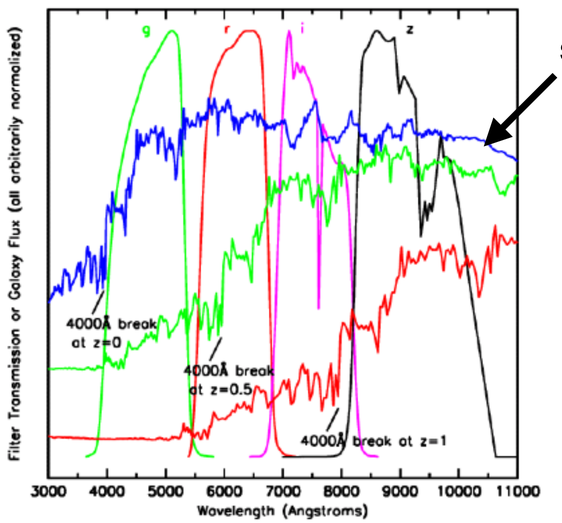


- We have images in multiple photometric large bands from ground based and space-based telescopes

- Using source detection softwares, we build catalogues of galaxies and measure their fluxes in each band in fixed angular apertures.

$\rightarrow \quad \text{data} \equiv \overrightarrow{\text{fluxes}}$ for each galaxy

SED fitting



Expected spectral energy distribution (SED) for a model galaxy

- Using physically motivated galaxy models, we can compute the expected observed fluxes for galaxies with known intrinsic properties $\overrightarrow{\psi}_{\text{phys}}$ (distance, stellar-mass $M_\star$, star-formation rate, etc.)

➔ We can thus solve the inverse problem in a bayesian framework through so-called *SED fitting* and get

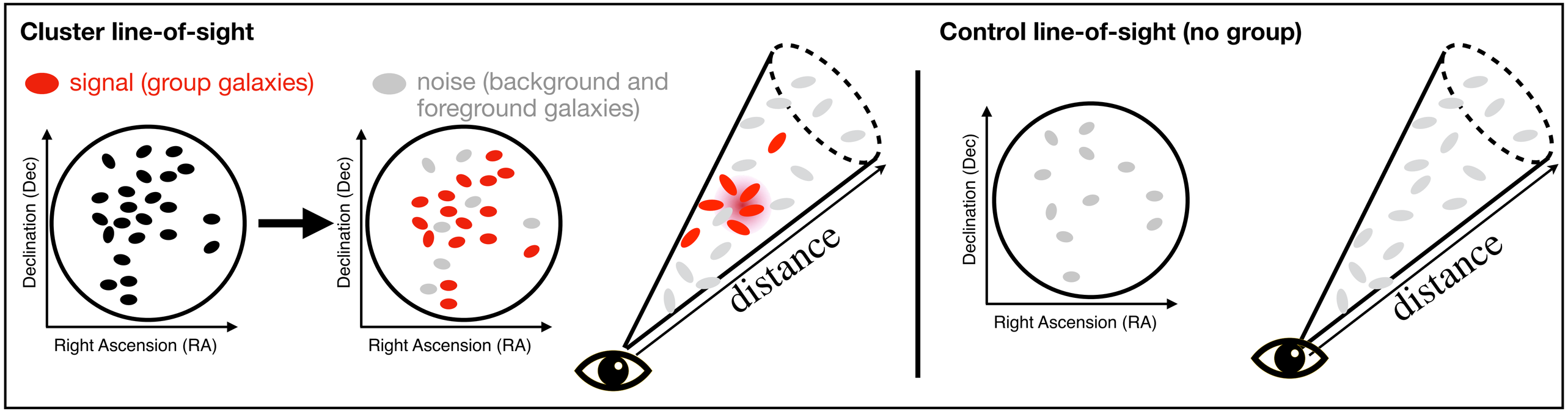
$$p(\overrightarrow{\psi}_{\text{phys}} \,|\, \text{data}) \propto p(\text{data} \,|\, \overrightarrow{\psi}_{\text{phys}})\; p(\overrightarrow{\psi}_{\text{phys}})$$

➔ In practice, my latest work focused on $p(M_\star \,|\, \overrightarrow{\text{fluxes}})$

Stellar-mass distribution in galaxy groups : data and model

When we look at galaxies nearby a galaxy group in photometric images, we do not know which galaxies belong to the group and which are foreground / background sources (noise).

But we can constrain the expected distribution of this noise by looking in a control line-of-sight where we know there is no galaxy group.



We model the observed galaxy stellar-mass distribution as the sum of signal + noise :

$$\phi_{\rm tot}(M_\star) = \delta_c \times \phi_{\rm signal}(\overrightarrow{\theta}^S_{\rm signal}) + \phi_{\rm noise}(\overrightarrow{\theta}^S_{\rm noise}) \times \frac{\Omega_{\rm clus}}{\Omega_{\rm control}}$$

e.g. $\phi(\overrightarrow{\theta}^S) \equiv$ Schechter distribution function

$\delta_c = 1$ if cluster, $\delta_c = 0$ if control

We take as the « observable » quantity the individual galaxy $M_\star$

But in fact we do not observe $M^i_\star$ but rather have a $p_i(M_\star \mid \overrightarrow{\text{fluxes}})$ from SED fitting